Name of the Program: **M. Tech (CSE)** Semester: **II**

Name of the Course: **Distributed Computing** Course Code: **MCS202**

Time: 1 Hour 30 Minutes Maximum Marks: 50

Note:

I. Answer **all the questions** by choosing ***any one of the sub questions***.

II. Each question carries 10 marks

Q1	(10 marks)	CO1
a)	(i)Identify motivations behind building Distributed System (DS)? What is the relationship between Parallel and Distributed Computing? Which are the other parallel computing approaches that can be adopted? Explain. (ii) Further, briefly discuss any one of examples of distributed system	
	OR	
b)	(i) In order to develop a distributed system, it is essential that every machine should have certain support from "Software Layers". Explain the role of various Software Layers with neat figure. (ii) Compare and contrast cloud computing with more traditional client-server computing? What is novel about cloud computing as a concept? What are the types of services offered by Cloud?	
Q2	(10 marks)	CO1
a)	Assessing Web-based services incur latency if every time the client contacts web-server. To reduce the latency organization should provision a facility that can cache frequently accessed WebPages. Explain with the help of a diagram the architecture that meets the mentioned requirement. Also provide the advantages and limitations	
	OR	
b)	"*Logical clock value assigned using Lamport's clock to two successive events in a process could differ by value* ***more than one***". Do you agree to this statement? Justify using time-space diagram of four processes and provide elaborate explanation using the appropriate **Implementation Rule**.	
Q3	(10 marks)	CO2
a)	What do you understand by mutual exclusion? Provide example. Identify the challenges that make mutual exclusion difficult in *Distributed Environment* as compared to processes on a single machine. Further, elaborate on the system model of distributed mutual exclusion.	
	OR	
b)	Consider Maekawa's algorithm. Show that the algorithm is deadlock prone. How does site suspect a deadlock? Provide detailed steps involved in handling deadlock and the messages involved.	
Q4	(10 marks)	CO2
a)	Consider four sites S1-S4 part of DS. Site S1, S2 and S3 intends to execute critical section. With the help of a Space-Time diagram, explain the steps involved in providing mutually exclusive access to critical section by using Ricart-Agrawala Algorithm	
	OR	
b)	Consider a Railway reservation system. A single seat needs to be allocated to only one traveler. There could be multiple traveler processes booking the seats. Propose a diagram depicting architecture where a single control process is nominated to handle requests from multiple traveler processes to ensure exclusive access to database.	
Q5	(10 marks)	CO3
a)	(i) What do you understand by the deadlock in collaborative computing processes? Explain with the help of a typical diagram. (ii) With reference to distributed deadlock, elaborate the assumptions and types of deadlocks.	
	OR	
b)	What are the advantages of Hierarchical Deadlock Detection Algorithms? Explain the Menasce and Muntz' hierarchical deadlock detection algorithm with the help of a neat diagram.	

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